

CS4865/6865

Assignment #2

Modelling data networks: selecting simulation parameters

Due date: October 9, 2025

Instructions: To be completed individually. Format your submission as a single pdf file and submit on D2L by the due date.

Background reading:

1. ns2 references from Lab 1
2. <https://en.wikipedia.org/wiki/Traceroute>

Question 1

- 1.1 Describe the `traceroute` Linux command.
- 1.2 Experiment with tracing routes to different servers on Internet (like cbc.ca or unb.ca). Pick a server which traceroute gives a (most) complete route for further analysis. Show this route (output from traceroute).
- 1.3 Using this output from traceroute estimate individual link delays along the route to the server from above. **Describe** how you calculated the individual link delays. (If any delay comes out negative, repeat Step 1.2 using a different destination server, **or use a very small number instead.**)
- 1.4 Research available tools for estimating the download speed to your computer (name a few tools, in references). Next, measure the download speed to your computer using one of these tools.

Question 2

- 2.1 Setup the simulation with the network nodes connected reflecting the route traced in Question 1, with individual link delays calculated in Step 1.3 above, and the link data rates set to the download speed measured for your computer (or greater) in Step 1.4. Set other simulation parameters as necessary for the simulation model of the network connection from Exercise 1. Submit the TCL code for this simulation (add some comments).
- 2.2 Run the simulation and show one screen shot showing the network forwarding packets.
- 2.3 Show the lines from the trace file that record the events of sending (h event), forwarding (multiple events (h events) along the route of the packet) and receiving (r event) at the final destination of the packet. **Discuss** how the values in the trace file are related (matching?) to the output from traceroute in Question 1. (Hint: the trace file time stamps are indicating one way delays, while the output from traceroute is the RTT. Is the packet (frame) transmission time accounted for in your modelling?)